



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : ME F314 / MF F314
Course Title : DESIGN OF MACHINE ELEMENTS
Instructor-in-Charge : Prof. MANI SANKAR DASGUPTA
Tutorial Instructors :

1. Course Description:

The course introduces students to concepts and nuances of Mechanical Engineering Design. Content includes materials, standards and design of machine elements and application of scientific principles, technical knowledge, and imaginative skills to be developed in a student. The machine elements covered are Shafts, Screw fasteners, Welded joints, Springs, Brakes & Clutches, Bearings and Gears.

The Tutorial classes of the course will be mainly used for assisted problem solving, consultation and evaluation.

2. Scope and Objective of the Course:

To prepare a student of mechanical engineering to apply theory and practice of Design of Mechanical Elements. It is an introductory course laying foundation on design ground rules, application of strength of material principles, selection of components, and selection of materials for a given application.

3. Text Books:

T1. Budynas, R. G. and Nisbett, K. J., "Shigley's Mechanical Engineering Design" Tata Mc-Grow Hill, 10th Edition, 2017 or a later ed.

4. Reference Books:

R1 Robert L Norton, Machine Design an integrated approach, fifth edition, Pearson Education Asia, 2018

R2 V B Bhandari, Design of Machine Elements, 5th edition Tata Mc-Grow Hill publishing Co, 2020

5. Course Plan:

Week no.	Module	Lect. no.	Refrence	Learning outcomes
1	Mechanical Engineering Design Principles, Preferred Number, Concept of Safety Factor, Engineering Units.	1-2	Ch1	To appreciate Design Philosophy, Understand problem formulation, Design considerations and Standards.



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Week no.	Module	Lect. no.	Reference	Learning outcomes
1	Material considerations in design and Material Specification.	3	Ch2	Ability to identify the important mechanical properties for particular application /loading conditions, selection of material.
2	Failure resulting from static loading	4-6	Ch5	To appreciate the relationship between strength and static loading in order to make decisions concerning size and shape for satisfying the requirements of safety and reliability.
3,4	Failure resulting from dynamic loading	7-10	Ch6	To appreciate the relationship between strength and dynamic/fatigue loading in order to make decisions concerning size and shape for satisfying the requirements of safety and reliability.
4,5	Shafts	11-14	Ch7	To understand the application of principle of mechanics to locate the critical areas and evaluate the stresses so as to meet the requirements of shaft and supported elements.
6	Screws fasteners & non-permanent joints	15-18	Ch8	Basic understanding of the mechanics of detachable joints, strength and stiffness estimation of joint.
7,8	Weldments and permanent joints	19-22	Ch9	Basic understanding of the permanent joints, strength and stiffness estimation of joint.
9	Mechanical springs	23-26	Ch10	Knowledge of various types of springs and their application and designing of spring
10	Rolling Contact bearings	27-30	Ch11	Basic understanding of various factors involved in selection of various types of anti-friction bearings.
11,12	Lubrication and Journal bearings	31-34	Ch12	Appreciation of need of lubrication and frictional heat dissipation, Journal bearing design and criteria
12,13	Clutches and Brakes	35-37	Ch16	Ability to analyse and design basic clutches and brakes.
14	Gear design fundamentals, Spur gear design	38-42	Ch13-15	To know about various types of gears and their applications. Appreciate various design features in gears. Application of conventional and AGMA design criteria, to iteratively decide gear material, surface treatment and dimensions for a Spur gear.

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Evaluative Tutorial*	30 Min	30	-	CB/OB
Mid-Semester Test	90 Min.	30	-	OB
Comprehensive Exam.	180 Min.	40	-	OB

* Best four performances in evaluated tutorials will be counted towards grading.



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7. OB: Open book, Prescribed Text Book only, clean book without any writing inside only will be permitted in examination, Books will be verified by invigilator during exam. No class notes or any other written material.

8. Chamber Consultation Hour: To be announced by individual instructors

9. Notices: Online at Nalanda

10. Make-up Policy: No Make-up is permissible in Tutorials (evaluated or non-evaluated). Make-up request for Mid Semester / Comprehensive examination must accompany appropriate supporting medical / exigency documents.

11. Criteria for NC: Below 10% of course topper's marks

12. Note (if any): Students to attend their registered sections only, tutorial attended in other section may not count towards evaluation.

Instructor-in-charge
Course No. ME F314 / MF F314